RESULT 1 / MEANING AND MODELING ERRORS

Only RIGOR adds meaning, with far fewer modeling errors.

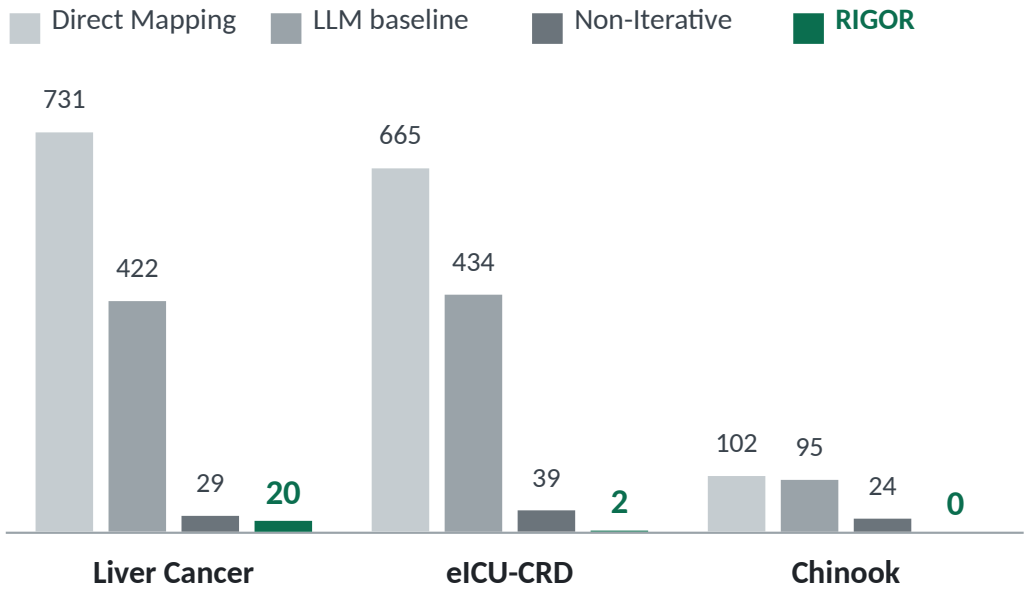
Ontologies generated with Claude for all three databases. Mistral and DeepSeek show the same pattern.

WHAT EACH METHOD ADDS

	Labels & comments	Provenance	Class hierarchy
Direct Mapping	○	○	○
LLM baseline	○	○	○
Non-Iterative	✓	○	○
RIGOR	✓	✓	✓

MODELING ERRORS FOUND BY OOPS!

fewer is better



Sources: paper Table 8 (labels, provenance, hierarchy in every ontology of each method) and Table 7 (OOPS! totals).

RESULT 2 / EXTERNAL BENCHMARK

RIGOR keeps the link to the data.

BURR benchmark (Laskowski et al., 2025) · ISWC scenario · highest F1 among the compared ontology-learning systems

WHAT BURR CHECKS

Every ontology element must map back to the database.

CLASSES	`person`	→	`Class: Person`
RELATIONS	`paper.author_id`	→	`ObjectProperty: author`
ATTRIBUTES	`person.name`	→	`DataProperty: name`

Elements match when they map to the same database elements, whatever their names. F1 per element type.

Illustrative elements · ISWC gold mapping: D2RQ example by Bizer & Seaborne

SYSTEM	CLASSES	RELATIONS	ATTRIBUTES
W3C-Mapper / Ontop rule-based	0.53	0.11	0.39
RDB2Onto rule-based	0.67	0.62	0.44
OntoGenix LLM-based	0.58	0.35	0.44
RIGOR LLM + rules	**0.73**	**0.67**	**0.52**

One scenario · 9 tables · 46 columns · 11 FKs · RIGOR with Claude · its prov:wasDerivedFrom links exported as a D2RQ mapping · Source: paper Table 13

RESULT 3 / LLM JUDGE AND HUMAN EXPERTS

An LLM judge and human experts both rank RIGOR first.

Both ranked four anonymized ontologies per table, from first to fourth.

LLM-AS-A-JUDGE · GPT-5.4

58 / 58

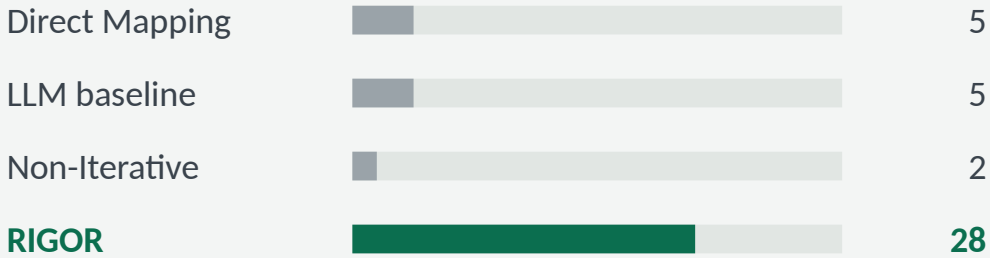
tables with RIGOR ranked first

Liver Cancer	**16 / 16**
eICU-CRD	**31 / 31**
Chinook	**11 / 11**

10 ONTOLOGY EXPERTS

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rankings with RIGOR first (70%)



LLM judge: 58 tables, 290 competency questions. Experts: 4 medical tables, significant agreement (Kendall's W 0.39–0.72). Sources: paper Tables 1 and 12.